Developing a Digital Maturity Model in the Apparel Industry

Website Requirement

I am planing to build the following maturity index calculation tool in vercer.

I am planing to have 2 pages

In the landing page of the website people can take start the survey

All the questions here , tiers and their weights should be configurable using a conf.yaml file . depending on the yaml file questions should appear in the website one after the other in an interactive way. Once the survey is completed users can go through the answers submitted and get their final scores and recommendations to improve the digital maturity of the organization

Users should be able to download their results and recommendations as a pdf. We will store the data in a json format but users might not able to retrieve a previously done survey

There should be another page explaining the theory behind the model which is a static page

<u>Aim and Objectives of this Study</u>

This study addresses the lack of a practical tool to quantify digital maturity apparel industry. The primary aim of the research is to develop a data-driven framework to measure digital maturity and demonstrate the potential advantages of digital transformation. This framework will support apparel manufacturers in transitioning toward sustainable, technology-driven business models. The study combines qualitative and quantitative data, including interviews with industry professionals to understand current practices, challenges, and strategic priorities, along with quantitative data from questionnaires. The study will be used to develop a model capable of indexing the level of digital transformation in apparel organizations, supporting them in enhancing their digital transformation journey.

<u>General Information</u>

1. Name:
2. Designation:
3. No. of years of Experience: Technical ______ Managerial ______
4. No of years of Experience in Digitalization ______
5. Highest level of Academic Qualification:

Instructions:

For each factor, select **one statement (1–5)** that best describes the current situation in your organisation.

Tier 1: Organisational Enablers

1. Leadership

Question:

To what extent does management support and drive digital transformation?

1 – Initial: Management provides little or no direction for digital transformation.

2 – Emerging: Management supports some digital initiatives when needed.

3 – Developing: Management has a clear digital vision and supports selected digital initiatives.

4 – Established: Management actively drives digital transformation, provides resources, and monitors progress.

5 – Advanced: Management continuously drives digital transformation, uses digital information for decision-making, and actively leads organisational change.

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2. Strategy & Governance

Question:
To what extent is digital transformation formally planned and managed in the organisation?

1 – Initial: There is no formal digital strategy, roadmap, or governance structure.

2 – Emerging: Digital initiatives are planned individually with limited formal guidance.

3 – Developing: The organisation has some digital priorities, policies, and governance practices.

4 – Established: Digital transformation is guided by a formal strategy, roadmap, governance structure, and regular reviews.

5 – Advanced: Digital strategy and governance are continuously reviewed and improved based on performance, risks, and emerging technologies.

3. People & Culture

Question:
To what extent are employees prepared and encouraged to support digital transformation?

1 – Initial: Employees have limited digital skills and there is significant resistance to digital change.

2 – Emerging: Basic digital training is provided to selected employees.

3 – Developing: Employees receive regular digital training and participate in some digital initiatives.

4 – Established: Continuous learning, employee participation, and cross-functional collaboration are well established.

5 – Advanced: Employees actively identify digital opportunities, develop new digital skills, and contribute to digital innovation.

4. Technology

Question:
To what extent are digital technologies and systems used and integrated across the organisation?

1 – Initial: Digital technologies are limited and mainly used for basic activities.

2 – Emerging: Some departments use digital tools, but systems are mostly separate.

3 – Developing: Digital systems are used across several functions with some level of integration.

4 – Established: Digital systems are well integrated and data is shared across relevant functions.

5 – Advanced: The organisation continuously integrates and improves its digital technologies using advanced analytics, AI, automation, and emerging technologies.

Tier 2: Core Operations

5. Research

Question:
To what extent are digital technologies and data used for product-related research?

1 – Initial: Research is mainly conducted using traditional methods with little use of digital tools.

2 – Emerging: Digital platforms or databases are occasionally used for research.

3 – Developing: Digital platforms and data are regularly used to support market, customer, and trend research.

4 – Established: Digital analytics and advanced tools are systematically used to identify trends, customer needs, and product opportunities.

5 – Advanced: AI, predictive analytics, and integrated data are continuously used to predict trends, customer preferences, and future product opportunities.

6. Design

Question:
To what extent are digital technologies used in the product design process?

1 – Initial: Product design is mainly conducted using traditional methods.

2 – Emerging: Basic digital design tools are used by selected employees or departments.

3 – Developing: Digital design and visualisation tools are regularly used, with some digital collaboration.

4 – Established: Digital design, visualisation, file sharing, and collaboration are integrated into the design process.

5 – Advanced: Advanced digital technologies such as 3D, AI, and virtual tools are continuously used for design optimisation, collaboration, and innovation.

7. Development

Question:

To what extent are digital technologies used to manage and improve product development?

1 – Initial: Product development is mainly conducted through physical and manual processes.

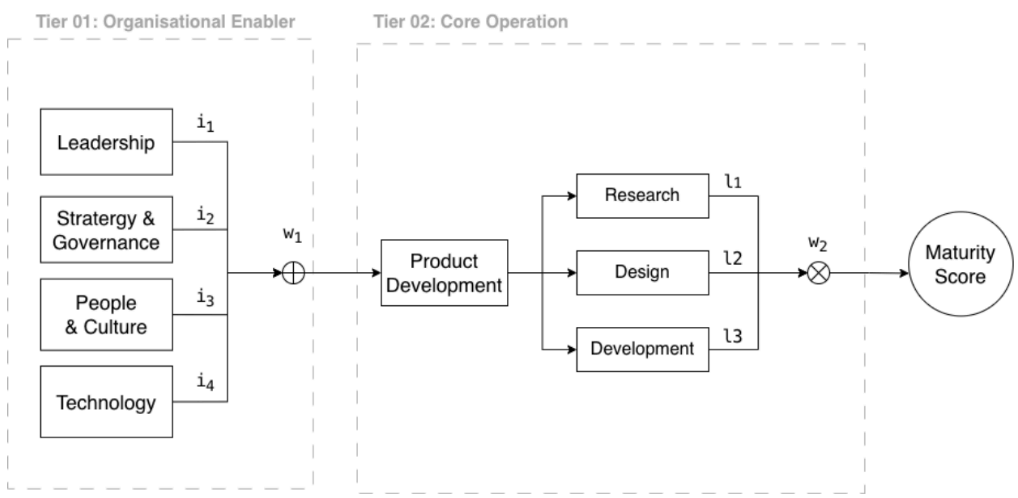
2 – Emerging: Individual digital tools are used for activities such as pattern development, sampling, or fit evaluation.

3 – Developing: Digital tools are regularly used for pattern development, 3D sampling, fit evaluation, or digital approval.

4 – Established: Digital tools, simulation, digital workflows, and integrated systems are used throughout product development.

5 – Advanced: Integrated systems, AI, analytics, simulation, and digital twins are used to continuously optimise product development and reduce time, cost, and physical sampling.

Proposed Model



On this based on the number they select it should multiply the weight
I1 = 0.383
I2 = 0.367
I3 = 0.190
I4 = 0.060

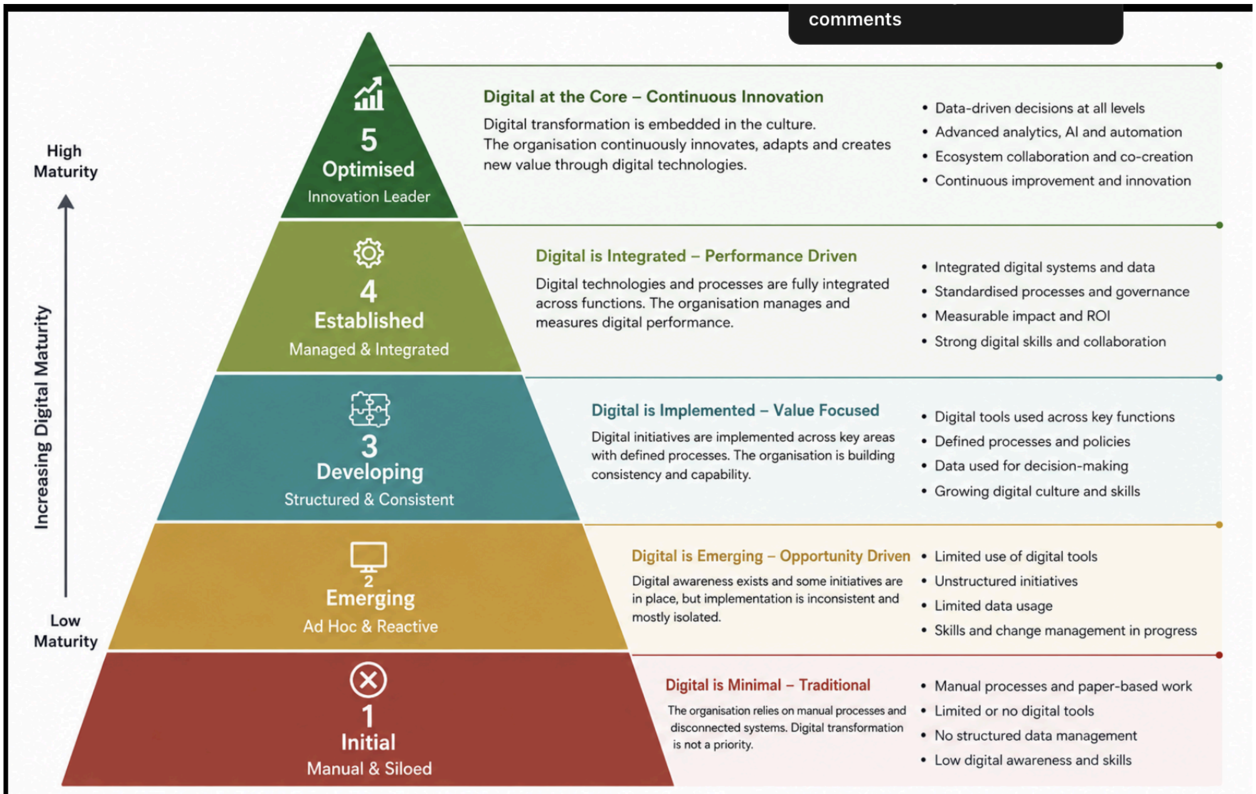
Sum = 0.383+0.367+0.190+0.060 = 1

L1 = 0.40
L2 = 0.23
L3 = 0.37

Sum =1

W1 =0.7
W2 = 0.3

Finally it should give the level of maturity



Maturity increases from Level 1 (Initial) to Level 5 (Optimised).